ADITYA RAJ

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SUMMARY

Software engineering student who builds and deploys web apps, including a live LLM-powered product, with hands-on work across frontend, serverless backend, LLM API integration, networking, and applied ML.
Hackathon team experience in backend and payment integration and in weather forecast verification. Develops with AI assistance and focuses on getting products working, tested, and live.

EDUCATION

VIT Bhopal University *2025 – 2029 (Expected)*
B.Tech in Computer Science and Engineering (AI & ML)
Coursework: Data Structures · OOP · Machine Learning · Distributed Systems · Networking · Discrete Mathematics · Operating Systems

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript
Frontend: React, Next.js, Vanilla JavaScript, Tailwind CSS, HTML, CSS, Responsive UI
Backend & Systems: Flask, FastAPI, Node.js, Express, REST APIs, Serverless (Vercel), TCP/IP Networking, Concurrency
Data & Services: Firebase, MongoDB, Razorpay
AI / ML: scikit-learn, NumPy, Pandas, NLP, Semantic Search, LLM API Integration
Security: API key isolation, defensive JSON parsing
Tools: Git, GitHub, VS Code, Vercel, Figma

FLAGSHIP PRODUCT — LIVE DEPLOYMENT

Baby Steps to the Mic — AI Public Speaking Coach *JavaScript · Web Speech API · Claude API · Vercel*
Built and deployed a live AI coaching web app with AI-assisted development. *Live demo* | *GitHub*

- Built a Vanilla JS frontend with Web Speech API integration for in-browser speech transcription and a generated speaking topic for each session
- Integrated the Claude API through a Vercel serverless backend to deliver structured per-session feedback: scores, strengths, weaknesses, and coaching recommendations
- Added defensive JSON parsing to safely handle malformed or unpredictable LLM responses without crashing the session
- Kept API keys server-side within serverless functions to prevent client-side exposure; handled environment configuration and deployment

PROJECTS

The More I Know, The Better — AI Learning Roadmap Builder *Next.js · React Flow · Anthropic SDK*
Live demo | *GitHub*

- Built an app that generates learning roadmaps with the Anthropic SDK and renders them as interactive node graphs
- Added a DAG integrity validator to check generated roadmaps, plus localStorage persistence for saved progress

Paperbrain — Focus & Discipline Tracker *React · Vite · Tailwind CSS · Framer Motion · Recharts*
Live demo | *GitHub*

- Developed a focus-tracking application featuring configurable Pomodoro timers, daily workload planning, and logged-session tracking
- Designed productivity analytics for focus endurance, procrastination patterns, consistency streaks, and peak productivity periods
- Implemented a 365-day activity heatmap and a discipline score based on consistency, focus endurance, and execution against planned targets

Distributed Chat System *Java · TCP/IP · Concurrency*

- Built a multi-client TCP socket server in Java with a dedicated thread per connection that routes messages across active client sessions
- Used thread synchronization to protect shared state and prevent race conditions under simultaneous writes

DocuQuery — AI Document Retrieval System *Python · NLP · IR*

- Built an NLP-based document querying pipeline with structured extraction and relevance-ranked retrieval across multi-format inputs
- Built a preprocessing and indexing layer to normalize, tokenize, and vectorize documents, with filtering and ranking logic for returned results

Offroad-Semantic — ML Semantic Analysis *Python · scikit-learn · NLP*

- Built a semantic classification pipeline covering text ingestion, cleaning, TF-IDF feature extraction, and scikit-learn model inference
- Designed semantic mapping workflows to assign structured category labels to unstructured inputs and compared multiple scikit-learn classifiers

PassSecure — Flask Password Security Tool *Python · Flask · JavaScript*

- Built a Flask web app that evaluates password strength using entropy estimation, pattern checks (dictionary words, sequences, repeated characters), and character class checks
- Handled input on both frontend and backend, with a responsive UI and a real-time strength indicator

EXPERIENCE

Core Member, Tech Team — The Matrix Club, VIT Bhopal *May 2026 – Present*

- Part of the tech team working on the club website

Core Member — Bit By Bit Club, VIT *Dec 2025 – Oct 2026*

- Helped organize Playtopia, a student gaming event with interactive stalls and activities
- Managed one of the game stalls and coordinated with participants and team members to keep the event running smoothly

Content Writer — InAmigos Foundation (IAF) *Apr 2026*

Graphic Designer — PranSpanda Club, VIT *Nov 2025 – Dec 2025*

- Designed posters and visual content for club activities and events, working on layout, typography, and branding

Event Anchor & Coordinator *2023 – 2024*

- Anchored and coordinated school and college events
- Developed public presentation and coordination skills under real-time constraints

HACKATHONS AND TEAM PROJECTS

Smart Canteen Queue & Pre-Order System — Team CypherX-HarshPal *Firebase · MongoDB · Razorpay*

- Collaborated with a team of five on a deployed canteen pre-order and queue-management platform built in an overnight hackathon
- Worked on backend integration with Firebase and MongoDB services and integrated Razorpay checkout (cards, UPI, net banking), with cash on delivery offered separately

TRINETRA — Smart India Hackathon 2026 (Team Ctrl ZZ) *Python · FastAPI · React*

- Team qualified the college round of SIH 2026; collaborating on a forecast-verification system that flags likely medium-range weather forecast failures and scores them from 0 to 100 across five weather parameters
- Identifies the contributing parameters and generates English and Hindi advisories; the prototype currently uses synthetic data, with live weather feed integration planned

Dawn of Code Hackathon — VIT (MERN Stack Club) *Participant*

- Built a full-stack prototype under time constraints

Smart VIT Hackathon *Participant*

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Industry Workshop — Amazon Q in Code & Play**
- **Machine Learning Foundations Certification**
- **Risk & Compliance in the Changing Cyber Ecosystem — Masterclass**
- **Deloitte Australia Data Analytics Job Simulation — Forage (Feb 2026):** built an interactive Tableau dashboard to analyze manufacturing downtime; applied Excel-based classification logic to categorize operational metrics
- **Deloitte Australia Cyber Security Job Simulation — Forage (Feb 2026):** analyzed 2,000+ lines of web activity logs to identify scripted API polling patterns and assess network exposure
- **Publication:** Mental Health Stigmas and Barriers to Help-Seeking Among Teenagers and Students in India (ResearchGate)
- **National-Level Financial Literacy Quiz Participant**
- **International Academic Engagement — VLU Global Talk**